Project Part #2: Establishing Requirements (10%)

Subject	:	Human-Computer Interaction Fundamentals (SCSM2113)
Session	:	2025/2026 Semester 1
Title	:	Project Part #2: Establishing Requirements
Duration	:	14 Days
Submission	:	This is a **TEAMWORK.** The team must upload the blog accordingly, by the 27th Nov 2025

NOTE TO STUDENTS

*This project continues from **Part 1 (Project Proposal)**, which explored the industry theme introduced by DreamEDGE — Digital Twin Visualisation Platform for Smart Industry. In this phase, you will deepen your understanding of users, their goals, and their interactions with digital twin systems to establish the **design requirements** for your proposed solution.*

*Your focus should remain on the **human experience**, i.e. how people interpret, interact with, and make decisions using digital twin systems, rather than on technical implementation. The goal is to define what the system should do **from the user's point of view** and identify usability and user experience requirements that will guide your interface design in the next phase.*

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Process 1: Identifying Needs and Establishing Requirements

Building on your approved proposal, you will now begin to materialize your solution by understanding your **target users**, their **tasks,** and the **context of use.** Designing an effective interaction requires knowing: - *Who your users are and what motivates them. - What tasks they perform and what challenges they face. - How they interpret system feedback and make decisions. - Under what conditions or environments they interact with the system.*

This process involves two main components: **User Analysis** and **Task Analysis**.

(a) User Analysis

Step 1: Identify Your Target Users

Refer back to the **three (3) target user groups** identified in your proposal. For the digital twin theme, your user groups could include (but are not limited to): Industrial Operators or Technicians – monitor and respond to real-time data from machines. Supervisors or Engineers – interpret performance metrics and make operational decisions. Trainees or Non-Technical Staff – learn from simulations and predictive data.

Step 2: Create Personas (3 Personas)

For each user group, create **one persona** (total of three personas). Each persona should include: - Background and role - Goals and motivations when using a digital twin interface - Skills and technological familiarity - Pain points (what frustrates them) - Gains or benefits (what they hope to achieve)

Use the **empathy map** approach to help you define what your user thinks, feels, says, and does.

Step 3: Define Tasks (3 Tasks)

Identify **three (3) key tasks** your proposed system should enable. At least two should be main functional tasks. Examples: 1. Monitor and interpret machine performance data. 2. Respond to alerts or predictive maintenance notifications. 3. Customise or filter dashboard visualisations.

For each persona, describe the context behind why they need to perform the task, their goals, and their expectations when doing it.

Step 4: Develop Scenarios (3 Scenarios)

Create one scenario for each persona that illustrates how they would use your system. Each scenario should include:

- The situation (context and environment of use)
- The user's goal or motivation
- The sequence of interaction with your system
- Expected outcome and how it helps them achieve their task

(b) Task Analysis and User Observation

To understand the real-world interaction flow, conduct a **Task Analysis** using the **Think Aloud** technique.

Step 1: Choose a Comparable Existing System

Find an existing digital twin platform or any dashboard/monitoring system with similar interaction patterns (for example, *Siemens MindSphere, Azure Digital Twins*, or any freely available demo/dashboard system). This will help you study how users currently perform tasks.

Step 2: Identify 3 Tasks

Use the same 3 tasks identified earlier (e.g., monitor data, interpret alerts, customise display).

Step 3: Observe 3 Real Users

Find **three (3)** users representing your personas. If you cannot find exact representatives, choose participants with similar profiles. Explain the think aloud method to them and ask them to complete the tasks using the existing system.

Record the sessions (screen + voice) as they perform each task, expressing their thoughts and reactions aloud. You should have **nine** (9) short videos (3 users × 3 tasks).

Step 4: Analyse Using Hierarchical Task Analysis (HTA)

For each task, construct an **HTA diagram** showing how users break down the task into subtasks. Create a total of nine **(9) HTAs** (one per user per task). At least one task should include multiple subtasks and decision points.

Step 5: Discuss and Derive Design Requirements

Compare how the three users perform the same task: - What difficulties or inefficiencies did you observe? - What feedback, visualization, or interface elements caused confusion? - Which steps could be simplified or automated? - What expectations were unmet?

From this comparison, extract **design requirements** for your proposed digital twin interface — for example: - The system should provide clear visual feedback for alerts. - Dashboard layout should minimise cognitive overload. - Predictive insights should be accompanied by explanation or confidence level.

Submission 2: Project Proposal

Your report should include the following sections:

Part 1: User Analysis

1. Title: Gathering Requirements - User Analysis
2. Proposed Tasks: List the three tasks of your proposed system.
3. Personas: Provide three distinct personas with two-paragraph descriptions and images.
4. Scenarios: Provide three corresponding task scenarios (one per persona) as two-paragraph narrations.

Part 2: Task Analysis

1. Title: Gathering Requirements - Task Analysis
2. Introduction: Describe briefly the existing system used for observation and list the three tasks.
3. Derivation of HTA: Include HTA diagrams and/or textual representations for each user performing each task (total of 9). Embed or link to corresponding videos.
4. Discussion: Discuss your findings and differences between users. Identify usability issues, insights from think-aloud, and lessons learned.
5. Design Requirements: Summarise the requirements your final design should fulfil based on the findings.

Reference Slides

Topics *05 Requirements and Data Gathering and 06- Persona and Tasks*

Scoring rubric:

Item	Absent (0)	Minimal (1)	Moderate (2)	Good (3)	Excellent (4)	Weight	Score
Persona	No persona identified	1–2 personas, incomplete details	3 personas, weak distinction	3 personas, mostly complete (profile, goals, pains, gains)	3 well-differentiated personas with full empathy mapping	2	/8
Scenario	No scenario provided	1–2 vague scenarios	3 scenarios missing some elements	3 detailed scenarios (persona + task + situation)	3 complete, realistic, and user-centred scenarios	2	/8
HTA	No HTA	Fewer than 6 HTAs, unclear	6–8 HTAs, mostly correct	9 HTAs, mostly accurate	9 HTAs, accurate and well-labelled	6	/24
Discussion	No discussion	Weak or missing insights	Some analysis, limited design link	Clear analysis, covers 2 of 3 points (HTA, differences, design)	Deep, reflective discussion linking all insights to design	3	/12
Videos	No video provided	<6 videos	6–8 videos	9 videos but unclear	9 clear, well-documented videos	3	/12

64/64 = 10%